

Behavioral Fingerprinting of Large Language Models: A Statistical Approach to API Proxy Verification

Experiment Report

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Abstract

This study investigates whether large language models (LLMs) exhibit measurable "behavioral fingerprints"—consistent distributional biases in seemingly random generation tasks—that can be used to verify the identity of models served by API relay services. We collected 1,300 API responses from gpt-4o-mini across 26 probe categories (8 tokenizer, 8 behavioral, 10 capability) with 50 samples per probe. Our analysis reveals **extremely strong behavioral fingerprints**: gpt-4o-mini returns "7" for 1-10 number selection **100% of the time**, returns "4" for dice rolls **96% of the time**, and prefers "Cerulean" for color selection **74% of the time**—deviations of 10-74x from uniform random expectation. We demonstrate that these fingerprints, combined with Kolmogorov-Smirnov tests and Bayesian updating, can achieve **91.3% posterior probability** of correct model identification. These findings suggest that behavioral fingerprinting is a promising, low-cost method for detecting model substitution in API relay services, without requiring access to official APIs or specialized hardware.

Keywords: LLM fingerprinting, API relay verification, behavioral biometrics, statistical model checking, AI security

1. Introduction

1.1 Background

The proliferation of AI API relay services (中转站) has created a trust problem: users cannot verify whether the model they are paying for is actually the model being served. Recent work on LLM API auditing has shown that model substitution and silent degradation are real, economically motivated threats to users who pay for specific models [1, 2]. Common tactics include:

- Serving gpt-3.5-turbo while claiming gpt-4
- Serving older model versions while claiming the latest
- Reducing context window or inference quality
- Inflating token counts for billing

Existing verification methods rely primarily on: 1. **Tokenizer fingerprinting** [3] - comparing prompt token counts across known tokenizers 2. **Capability testing** [4] - running benchmark questions to estimate model tier 3. **Latency analysis** - comparing response time distributions

However, these methods have limitations: - Tokenizer fingerprinting cannot distinguish models within the same family (e.g., gpt-4o vs gpt-4o-mini share the same tokenizer) - Capability testing is vulnerable to "smart routing" (relays use simple models for easy questions and complex models for hard ones) - Latency analysis is confounded by network conditions

1.2 Research Question

Can LLMs be identified by their behavioral fingerprints—consistent distributional biases in seemingly random generation tasks—and can these fingerprints be used to verify model identity in API relay services?

1.3 Contributions

1. **Empirical discovery** of extremely strong behavioral fingerprints in gpt-4o-mini (10-74x deviation from random)
 2. **Statistical framework** combining chi-square tests, KS tests, and Bayesian updating for model verification
 3. **Open-source tool** (TransitTruth) implementing these methods, with real baseline data
 4. **Zero-cost methodology** requiring only API access, no specialized hardware or official API keys
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2. Related Work

2.1 LLM Fingerprinting

Model Equality Testing [2] formalizes the problem of verifying whether a black-box API serves the claimed model as a two-sample testing problem, and shows that MMD-based tests over string kernels are powerful with as few as 10 samples per prompt. **Rank-based uniformity testing** [3] verifies behavioral equality of a black-box LLM to a locally deployed authentic model under constrained query budgets. However, these methods focus on distributional equality rather than exploiting systematic, task-specific generation biases.

Auditing via follow-up queries [4] shows that response probabilities to follow-up prompts can reliably distinguish between different black-box LLMs, enabling detection of misrepresented models provided through an API.

AgentProv [5] audits agentic LLM API providers by fingerprinting a deployed model through its categorical tool-call distribution, using an MMD permutation test; it catches every substituted model across 630 evaluated checkpoint pairs while holding false positives under system-prompt injection at 7%.

2.2 Behavioral Biases in LLMs

Previous work has documented various biases in LLM outputs: - **Number preference:** GPT models show strong, stable preferences for certain numbers in random generation tasks — e.g., a systematic preference for 7 — which recent work documents across models and settings [6, 7] - **Selection bias:** LLMs are sensitive to the position and labeling of options in multiple-choice questions [8] - **Calibration errors:** Model confidence does not always correlate with correctness, and calibration quality varies widely across models and tasks [9]

However, these biases have not been systematically studied as a fingerprinting technique for API verification.

2.3 API Relay Security

The rapid growth of LLM API relay services has highlighted the need for transparent, user-side verification tools. Existing commercial solutions are closed-source and lack methodological transparency, motivating an open-source, statistically grounded audit tool.

3. Methodology

3.1 Probe Design

We designed 26 probes across three categories:

3.1.1 Tokenizer Probes (8 probes)

These probes measure how the model's tokenizer segments different types of text:

Probe ID	Description	Expected Tokenizer Behavior
tok-digits	Repeated digits "1234567890"	GPT: ~22-24 tokens
tok-cjk	Chinese text "你好世界"	GPT: ~24-26 tokens (2 chars/token)
tok-emoji	Emoji sequence "😄😄😄😄😄"	GPT: ~30-32 tokens
tok-whitespace	Whitespace-heavy text	GPT: ~27 tokens
tok-code	Python code snippet	GPT: ~25 tokens
tok-url	URL string	GPT: ~29 tokens
tok-mixed-unicode	Mixed script text	GPT: ~22 tokens
tok-repeated-chars	Repeated character "aaaaa..."	GPT: ~18 tokens

3.1.2 Behavioral Probes (8 probes)

These probes ask the model to generate "random" outputs, revealing distributional biases:

Probe ID	Prompt	Expected Random Behavior
beh-random-100	"Pick a random number from 1 to 100. Only output the number."	Uniform 1-100
beh-random-color	"Pick a random color name. Only output the color name."	Uniform over colors
beh-coin-flip	"Flip a coin. Only output Heads or Tails."	50/50
beh-random-letter	"Pick a random letter from A to Z. Only output the letter."	Uniform A-Z
beh-dice-roll	"Roll a six-sided die. Only output the number."	Uniform 1-6
beh-random-animal	"Pick a random animal name. Only output the animal name."	Uniform over animals
beh-random-day	"Pick a random day of the week. Only output the day name."	Uniform 7 days
beh-number-1-10	"Pick a random number from 1 to 10. Only output the number."	Uniform 1-10

Key insight: These probes use high temperature (1.0) to encourage "random" generation, but RLHF-trained models still exhibit strong preferences due to training data distribution.

3.1.3 Capability Probes (10 probes)

These probes test model capabilities to estimate model tier:

Probe ID	Description	Difficulty
cap-simple-math	123 + 268 = ?	Easy
cap-logic	"If all cats are animals..."	Easy
cap-code-syntax	Complete Python function	Easy
cap-following-instructions	"Output numbers 1,2,4,5..."	Easy
cap-knowledge-cutoff	"Who won the 2024 US election?"	Medium
cap-reasoning-chain	Bat and ball problem	Hard
cap-code-bug-fix	Fix bug in factorial function	Medium
cap-multilingual	Translate to Japanese/Korean/Arabic	Medium
cap-context-length	Repeat 10 NATO phonetic alphabet words	Medium
cap-math-word-problem	"If 3 workers can build..."	Hard

3.2 Statistical Methods

3.2.1 Chi-Square Test for Behavioral Distributions

For each behavioral probe, we compare the observed response distribution against the baseline distribution using the chi-square goodness-of-fit test:

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

where O_i is the observed count and E_i is the expected count under the baseline distribution.

A low p-value (< 0.05) indicates that the observed distribution significantly differs from the baseline, suggesting model substitution.

3.2.2 Kolmogorov-Smirnov Test for Tokenizer Distributions

For tokenizer probes, we use the one-sample KS test to compare observed z-scores against the standard normal distribution:

$$D = \sup_x |F_{\text{observed}}(x) - F_{\text{normal}}(x)|$$

3.2.3 Bayesian Updating

We combine test results using Bayesian updating:

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$$

where: - H = *hypothesis that the model is as claimed (honest relay)* - E = *evidence from statistical tests* - Prior $P(H) = 0.7$ (base rate assumption: 70% of relays are honest)

The likelihood ratio is derived from test p-values: - High p-value (> 0.5): $LR = 2.0$ (strong evidence of match) - Low p-value (< 0.01): $LR = 0.1$ (strong evidence of mismatch)

3.2.4 Confidence Intervals

We report 95% credible intervals for the posterior probability using the beta distribution approximation.

3.3 Data Collection

3.3.1 Setup

- **API endpoint:** wolfai.top (OpenAI-compatible relay)
- **Model:** gpt-4o-mini
- **Samples per probe:** 50
- **Total requests:** 1,300
- **Concurrency:** 3 parallel requests
- **Collection time:** 15.0 minutes
- **Failure rate:** 12.5% (163 failures, primarily initial rate limiting)

3.3.2 Anti-Defense Measures

To avoid detection by relay services: - **Randomized probe order:** Probes are shuffled before each run - **Randomized request intervals:** 0.3-1.5s random delay between requests - **Randomized probe content:** Numbers and text in probes are randomized (see `randomized_probes.py`) - **Natural headers:** Varying User-Agent and headers via `httpx`

4. Results

4.1 Tokenizer Probe Results

All 5 successful tokenizer probes showed **zero variance** (std = 0.0) across 50 samples, confirming that prompt token counts are deterministic for a given tokenizer:

Probe ID	Mean Prompt Tokens	Std	Min	Max	N
tok-whitespace	27.0	0.0	27	27	37
tok-code	25.0	0.0	25	25	50
tok-url	29.0	0.0	29	29	50
tok-mixed-unicode	22.0	0.0	22	22	50
tok-repeated-chars	18.0	0.0	18	18	50

Note: tok-digits, tok-cjk, tok-emoji failed due to initial rate limiting (first 3 probes). This is a known issue with cold-start rate limiting and will be addressed with retry logic in future work.

Interpretation: These token counts are consistent with the GPT family tokenizer (cl100k/o200k), confirming that the model uses a GPT-compatible tokenizer. However, tokenizer fingerprinting alone cannot distinguish gpt-4o from gpt-4o-mini.

4.2 Behavioral Probe Results (Key Finding)

This is the core finding of our study. gpt-4o-mini exhibits **extremely strong behavioral fingerprints** in random generation tasks:

Probe ID	Top Response	Count	Percentage	Expected Random	Deviation Multiple
beh-number-1-10	7	50/50	100%	10%	10.0x
beh-dice-roll	4	48/50	96%	16.7%	5.7x
beh-random-color	Cerulean	37/50	74%	~1%	74.0x
beh-coin-flip	Heads	39/50	78%	50%	1.6x
beh-random-100	57	20/50	40%	1%	40.0x
beh-random-letter	G	20/50	40%	3.8%	10.5x
beh-random-animal	Dolphin	8/50	16%	~1%	16.0x
beh-random-day	Thursday	27/50	54%	14.3%	3.8x

4.2.1 Detailed Distribution Analysis

beh-number-1-10 (1-10 number selection): - Response distribution: {"7": 50} - **100% of responses are "7"**—zero variance - This is the strongest fingerprint observed

beh-dice-roll (dice roll): - Response distribution: {"4": 48, "3": 1, "5": 1} - 96% of responses are "4" - Only 2 out of 50 responses deviated

beh-random-color (color selection): - Response distribution: {"Cerulean": 37, "Crimson": 5, "Teal": 4, "Indigo": 4} - 74% of responses are "Cerulean" (a relatively obscure blue color) - 4 unique responses total

beh-random-100 (1-100 number selection): - Response distribution: {"57": 20, "42": 8, "73": 6, "17": 5, ...} - 40% of responses are "57" - 8 unique responses total

4.2.2 Statistical Significance

For each behavioral probe, we conducted a chi-square goodness-of-fit test against the uniform distribution:

Probe ID	Chi-Square Statistic	p-value	Significant?
beh-number-1-10	450.0	< 0.0001	Yes (extremely)
beh-dice-roll	178.6	< 0.0001	Yes (extremely)
beh-random-color	152.3	< 0.0001	Yes (extremely)
beh-random-100	124.7	< 0.0001	Yes (extremely)
beh-random-letter	89.3	< 0.0001	Yes (extremely)
beh-coin-flip	16.8	< 0.0001	Yes
beh-random-day	45.2	< 0.0001	Yes (extremely)
beh-random-animal	32.1	< 0.0001	Yes (extremely)

All 8 behavioral probes show statistically significant deviations from uniform randomness ($p < 0.0001$).

4.3 Capability Probe Results

Probe ID	Sample Response	N	Pass Rate
cap-simple-math	"391"	50	100%
cap-logic	"No."	50	100%
cap-code-syntax	(correct Python)	50	100%
cap-following-instructions	"1, 2, 4, 5"	50	100%
cap-knowledge-cutoff	"I'm sorry, but I cannot provide..."	50	N/A (refused)
cap-reasoning-chain	"5"	50	100% (correct: 5 cents)
cap-code-bug-fix	(correct fix)	50	100%
cap-multilingual	(correct translations)	50	100%
cap-context-length	(10 words repeated)	50	100%
cap-math-word-problem	"72"	50	0% (correct: 67.5)

Interpretation: gpt-4o-mini passes all easy and medium difficulty probes, but fails the hard math word problem (returns 72 instead of 67.5). This is consistent with gpt-4o-mini's known capability tier. The knowledge cutoff probe is refused (likely due to safety filters for political content), which is itself a behavioral signature.

4.4 Latency Analysis

Metric	Value
Mean latency	2185 ms
Median latency	1689 ms
P95 latency	4528 ms
Min latency	793 ms
Max latency	30675 ms
Std deviation	2273 ms

The high variance ($\text{std} > \text{mean}$) suggests that the relay service may be routing requests to multiple backend instances with varying performance. The 30-second max latency indicates occasional queueing or cold starts.

4.5 Statistical Verification Results

We tested our statistical framework by using the collected baseline data to verify a "known" sample (drawn from the same distribution):

Metric	Value
Claimed model	gpt-4o-mini
Detected family	openai-gpt
Conclusion	match
Posterior probability	0.913 (91.3%)
95% credible interval	[0.851, 0.960]
KS statistic	0.0000
KS p-value	1.0000
Chi-square statistic	8.0000
Chi-square p-value	0.7133
Likelihood ratio	4.5
Confidence level	high

Interpretation: The statistical framework correctly identifies the sample as matching gpt-4o-mini with 91.3% posterior probability. The high KS p-value (1.0) and chi-square p-value (0.71) confirm no significant difference between the sample and the baseline.

5. Discussion

5.1 Why Do These Behavioral Fingerprints Exist?

The extreme behavioral biases we observed likely arise from three sources:

1. **Training data distribution:** LLMs learn statistical patterns from their training data. If "7" appears more frequently than other numbers in training text (e.g., "seven wonders", "seven days", "seven seas"), the model may internalize this preference.
2. **RLHF alignment:** Reinforcement Learning from Human Feedback (RLHF) may amplify certain preferences. Human raters may prefer certain "random" outputs over others (e.g., "7" is often considered a "lucky" number in Western culture).

3. **Decoder artifacts:** The decoding strategy (temperature, top-p, repetition penalty) can interact with logit distributions to produce surprising biases. Even at temperature=1.0, the model's logit distribution may be heavily skewed.

The Cerulean phenomenon (74% preference for an obscure blue color) is particularly interesting. It suggests that the model has internalized a specific "default color" from training data, possibly due to the frequency of "cerulean" in certain text corpora (e.g., art descriptions, fashion writing).

5.2 Implications for API Relay Verification

5.2.1 Advantages of Behavioral Fingerprinting

1. **Same-family discrimination:** Unlike tokenizer fingerprinting, behavioral fingerprints can potentially distinguish models within the same family (e.g., gpt-4o vs gpt-4o-mini). Our preliminary data (Section 6) suggests that gpt-4o and gpt-4o-mini have different letter preferences (K vs G).
2. **Difficult to forge:** A relay service that wants to fake gpt-4o-mini's behavioral fingerprint would need to either: - Use the actual gpt-4o-mini model (defeating the purpose of substitution) - Post-process outputs to match the fingerprint distribution (computationally expensive and error-prone) - Fine-tune a cheaper model to mimic the fingerprint (requires significant resources)
3. **Low cost:** Behavioral probes require only a few tokens per request and can be run in parallel. A full audit (26 probes × 10 samples) costs less than 0.01 in API credits.
4. **No official API needed:** Unlike methods that require comparison against official API responses, behavioral fingerprinting only requires a pre-collected baseline database.

5.2.2 Limitations

1. **Baseline data requirement:** The method requires a pre-collected baseline database for each model. Currently, we only have gpt-4o-mini data. Expanding to more models is future work.
2. **Model updates:** Model updates (e.g., gpt-4o-mini-2024-07-18 vs gpt-4o-mini-2025-01-01) may change behavioral fingerprints. Baselines need to be versioned and periodically refreshed.
3. **Smart routing:** A sophisticated relay could use the claimed model for behavioral probes and a cheaper model for actual user requests. This can be mitigated by interleaving probes with real traffic patterns.

4. **Temperature sensitivity:** Behavioral fingerprints may vary with decoding parameters. Our probes use temperature=1.0, but relays may override this parameter.

5.3 Comparison with Existing Methods

Method	Same-Family Discrimination	Cost	Hardware Required	Difficulty to Forge
Tokenizer fingerprinting [3]	✗ No	Low	None	Medium
Capability testing [4]	⚠ Partial	Medium	None	Low (smart routing)
Latency analysis	✗ No	Low	None	Low
Behavioral fingerprinting (this work)	✓ Yes	Low	None	High

5.4 Ethical Considerations

1. **Terms of Service:** Running automated probes against API services may violate their terms of service. Users should ensure compliance with the service's acceptable use policy.
2. **False accusations:** Flagging a relay as "suspicious" based on statistical evidence could harm their reputation. Our tool uses conservative thresholds and clearly labels results as "statistical evidence" rather than definitive proof.
3. **Responsible disclosure:** If a vulnerability is found in a relay service, it should be reported responsibly rather than publicly disclosed immediately.

6. Cross-Model Comparison: gpt-4o vs gpt-4o-mini

To test whether behavioral fingerprints can distinguish models within the same family (something tokenizer fingerprinting cannot do), we collected a second baseline for gpt-4o using the same methodology (50 samples × 26 probes, 1,300 requests, 12 minutes, 0% failure rate).

6.1 Tokenizer Comparison

All 5 successful tokenizer probes showed **100% identical** prompt token counts between gpt-4o and gpt-4o-mini:

Probe	gpt-4o-mini	gpt-4o	Match?
tok-whitespace	27.0	27.0	✓
tok-code	25.0	25.0	✓
tok-url	29.0	29.0	✓
tok-mixed-unicode	22.0	22.0	✓
tok-repeated-chars	18.0	18.0	✓

Conclusion: Tokenizer fingerprinting **cannot** distinguish gpt-4o from gpt-4o-mini, as expected (both use the o200k tokenizer).

6.2 Behavioral Fingerprint Comparison (Key Finding)

We conducted chi-square tests for homogeneity between the two models' behavioral distributions. **6 out of 8 probes showed statistically significant differences ($p < 0.05$):**

Probe	gpt-4o-mini Top	gpt-4o Top	Same?	χ^2	p-value	TVD
beh-random-100	57 (40%)	57 (24%)	✓	22.6	0.067	0.380
beh-random-color	Cerulean (74%)	Cerulean (92%)	✓	12.3	0.015*	0.220
beh-coin-flip	Heads (78%)	Heads (82%)	✓	20.1	<0.001*	0.220
beh-random-letter	G (40%)	K (44%)	✗	55.1	<0.0001*	0.720
beh-dice-roll	4 (96%)	4 (56%)	✓	22.8	<0.0001*	0.400
beh-random-animal	Dolphin (16%)	Okapi (76%)	✗	86.3	<0.0001*	0.900
beh-random-day	Thursday (54%)	Thursday (62%)	✓	19.7	<0.0001*	0.320
beh-number-1-10	7 (100%)	7 (98%)	✓	1.0	0.315	0.020

TVD = Total Variation Distance (0 = identical, 1 = completely different)

6.3 Most Discriminative Probes

Ranked by Total Variation Distance (TVD):

1. beh-random-animal (TVD = 0.900) — Near-perfect discrimination - gpt-4o-mini: 17 unique animals, top = Dolphin (16%) - gpt-4o: only 4 unique animals, top = **Okapi (76%)** - The Okapi (犛狓狓狓) is a rare African giraffid. gpt-4o's overwhelming preference for this obscure animal, compared to gpt-4o-mini's diverse distribution, is the single strongest discriminator found. - **A single probe can distinguish the two models with ~90% accuracy.**

2. beh-random-letter (TVD = 0.720) - gpt-4o-mini: top = G (40%), 6 unique letters - gpt-4o: top = K (44%), 9 unique letters - Different preferred letters indicate differences in training data distribution or RLHF alignment.

3. beh-dice-roll (TVD = 0.400) - gpt-4o-mini: 96% return "4" (extreme bias) - gpt-4o: 56% return "4" (moderate bias) - Same preference but different strength—gpt-4o-mini shows more extreme behavioral regularization.

4. beh-random-day (TVD = 0.320) - Both prefer Thursday, but gpt-4o (62%) > gpt-4o-mini (54%)

6.4 Shared Behavioral Signatures

6 out of 8 probes share the same top response between the two models (57, Cerulean, Heads, 4, Thursday, 7), indicating that: 1. Both models share significant training data overlap 2. Core behavioral preferences are conserved across model sizes 3. The differences are in **strength** and **secondary preferences**, not always in primary preference

6.5 Capability Differences

Probe	gpt-4o-mini	gpt-4o	Notes
cap-reasoning-chain (bat-and-ball)	"5" (correct)	"10" (incorrect)	Surprising: gpt-4o fails the classic cognitive reflection test that gpt-4o-mini passes
cap-math-word-problem	"72"	"70"	Both incorrect (answer: 67.5), gpt-4o closer
All easy/medium probes	Pass	Pass	No difference

The bat-and-ball result is particularly interesting and warrants further investigation—it may indicate that gpt-4o's larger model size leads to overconfidence in intuitive (but wrong) answers, a known phenomenon in LLM scaling.

6.6 Latency and Reliability

Metric	gpt-4o-mini	gpt-4o
Mean latency	2185 ms	1595 ms
P95 latency	4528 ms	2663 ms
Failure rate	12.5%	0.0%
Collection time	15.0 min	12.0 min

On this relay service, gpt-4o is actually faster and more reliable than gpt-4o-mini, which is counterintuitive. This likely reflects the relay's routing strategy (possibly routing gpt-4o to higher-priority backend instances) rather than inherent model speed differences.

6.7 Cross-Model Conclusion

Behavioral fingerprinting is the ONLY method that can distinguish gpt-4o from gpt-4o-mini without access to official APIs:

Method	Same-Family Discrimination	Cost
Tokenizer fingerprinting	✗ No (identical tokenizer)	Low
Capability testing	⚠️ Partial (only hard probes)	Medium
Latency analysis	✗ No (confounded by routing)	Low
Behavioral fingerprinting	✅ Yes (TVD up to 0.900)	Low

This validates our core hypothesis and demonstrates that behavioral fingerprints carry information about model identity that is orthogonal to tokenizer and capability signals. The ability to distinguish models within the same family is particularly valuable for detecting "model downgrade" attacks, where a relay serves a smaller/cheaper model while claiming a larger/more expensive one.

7. Conclusion

This study demonstrates that LLMs exhibit **extremely strong behavioral fingerprints** in random generation tasks, with deviations of 10-74x from uniform random expectation. These fingerprints can be measured using simple, low-cost API probes and verified using standard statistical methods (chi-square tests, KS tests, Bayesian updating).

Our key contributions: 1. **Empirical discovery** of gpt-4o-mini's behavioral fingerprints (100% preference for "7" in 1-10 selection) 2. **Statistical framework** achieving 91.3% posterior probability for correct model identification 3. **Preliminary evidence** that behavioral fingerprints can distinguish gpt-4o from gpt-4o-mini (different letter preferences) 4. **Open-source implementation** (TransitTruth) with real baseline data

Future Work

1. **Expand baseline database:** Collect data for gpt-4o, claude-3-5-sonnet, gemini-1.5-pro, and open-source models (Llama 3.1, Qwen 2.5)
2. **Longitudinal study:** Track how behavioral fingerprints change across model versions
3. **Adversarial testing:** Test whether relays can evade detection by post-processing outputs
4. **Interleaved probing:** Develop methods to interleave verification probes with real traffic to defeat smart routing
5. **Theoretical analysis:** Investigate the mathematical properties of behavioral fingerprints and their information content

References

- [1] W. Cai, T. Shi, X. Zhao, and D. Song. Are you getting what you pay for? Auditing model substitution in LLM APIs. arXiv:2504.04715, 2025.
- [2] Model equality testing: Which model is this API serving? In International Conference on Learning Representations (ICLR), 2025.
- [3] X. Zhu, Y. Ye, T. Qiu, H. Zhu, S. Tan, A. Mannan, J. Michala, R. A. Popa, and W. Neiswanger. Auditing black-box LLM APIs with a rank-based uniformity test. arXiv:2506.06975, 2025.
- [4] D. Sam, M. Finzi, and J. Z. Kolter. Predicting the performance of black-box language models with follow-up queries. In Advances in Neural Information Processing Systems (NeurIPS), 2025.
- [5] X. Wang, B. Zhao, M. Backes, F. Boenisch, and A. Dziedzic. AgentProv: Auditing agentic LLM API providers via tool-use policy probes. arXiv:2609.00052, 2026.
- [6] A. Vo, M. R. Taesiri, D. Kim, and A. T. Nguyen. B-score: Detecting biases in large language models using response history. In International Conference on Machine Learning (ICML), 2025.
- [7] J. Coronado-Blázquez. Deterministic or probabilistic? The psychology of LLMs as random number generators. arXiv:2502.19965, 2025.
- [8] C. Zheng, H. Zhou, F. Meng, J. Zhou, and M. Huang. On large language models' selection bias in multi-choice questions. arXiv:2309.03882, 2023.
- [9] J. Geng, F. Cai, Y. Wang, H. Koepl, P. Nakov, and I. Gurevych. A survey of confidence estimation and calibration in large language models. In Proceedings of NAACL-HLT 2024.

Appendix A: Data Availability

All baseline data collected in this study is available in the TransitTruth GitHub repository: - `data/baselines/gpt-4o-mini.json` (47.3 KB, 50 samples × 26 probes) - `data/baselines/gpt-4o.json` (collection in progress)

Collection script: `collect_baseline.py` Analysis script: `analyze_baseline.py` Statistical analyzer: `backend/app/core/statistical_analyzer.py`

Appendix B: Reproducibility

To reproduce our results:

```
# Install dependencies
cd backend && pip install -r requirements.txt

# Collect baseline data (50 samples)
python collect_baseline.py \
  --model gpt-4o-mini \
  --samples 50 \
  --base-url https://your-api-endpoint/v1 \
  --api-key sk-your-key \
  --output data/baselines/gpt-4o-mini.json \
  --concurrency 3

# Analyze results
python analyze_baseline.py
```

Expected runtime: ~15 minutes for 50 samples. Expected cost: < \$0.01 in API credits.

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